The Lemke Oliver–Soundararajan Bias as Surface Evidence of Ternary Constitutional Structure

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This paper was not planned. On the Friday afternoon before submitting the Constitutional Forcing programme to peer review, the author read a popular account of the Lemke Oliver–Soundararajan prime digit bias—a result that has been described as "deeply unsettling" since its publication in 2016. The explanation appeared immediately, in a single conversation, without calculation. Constitutional Forcing was already in the room. The bias recognized the framework. This paper is the formalization of what took approximately one afternoon to see.

Abstract

In 2016, Lemke Oliver and Soundararajan published a striking computational discovery: the last digits of consecutive primes exhibit a strong repulsion bias that contradicts naïve randomness assumptions and persists—though diminishing—across all number bases tested. The authors correctly traced this to the Hardy–Littlewood prime constellations conjecture, but could not identify the underlying mechanistic cause: why the structure exists, why it holds across all bases, and why the diminishment rate is precisely what it is. We demonstrate that the bias is not mysterious: it is the direct base-10 projection of ternary constitutional pressure governing prime distribution. Specifically, the observed digit-repulsion is a shadow of the mod 3 constraint under which all primes (except 2 and 3) reside in remainder classes $\{1, 2\}$, and under which all twin prime pairs satisfy $p \equiv 2 \pmod 3$. The *snail's pace* of diminishment is identified as the Siegel–Walfisz convergence rate of ternary class equilibration—not an empirical curiosity, but a fundamental quantity in analytic number theory appearing in a context where it was not previously recognized. This explanation is mechanistic rather than descriptive: it does not merely reproduce the bias from the Hardy–Littlewood conjecture, but identifies the structural reason the conjecture itself encodes such behaviour.

Keywords: prime digit bias, ternary constitutional structure, Lemke Oliver–Soundararajan, mod 3 constraints, minimum gap structure, twin primes, parity barrier, Siegel–Walfisz theorem, cascade moduli

MSC2020: 11N05 (primary), 11N13, 11A41, 94A17 (secondary)

1 Introduction: The Mystery That Wasn't

In 2016, the mathematical world was surprised by a result so elementary that any undergraduate could verify it in an afternoon—and yet no one had checked.

If primes were distributed randomly among their possible terminal digits, each of the four digits $\{1, 3, 7, 9\}$ should follow any other with probability approximately 25%. This is what randomness means. Lemke Oliver and Soundararajan tested the first billion primes [LOS16]. The answer violated the assumption.

A prime ending in 9 is followed by a prime ending in 1 roughly 65% more often than by another prime ending in 9. Every prime actively avoids repeating its own terminal digit. The bias holds across every number base tested. It diminishes—but at what the authors themselves called a snail's pace.

They correctly traced its theoretical origin to the Hardy–Littlewood prime constellations conjecture [HL23]. But this explanation is descriptive rather than mechanistic. It says that the Hardy–Littlewood conjecture, if true, implies the bias. It does not explain:

(i) Why the structure exists in the first place;

(ii) Why it holds across all number bases, not just base 10;

(iii) Why the diminishment rate is precisely what it is;

(iv) What the bias is a symptom of at a deeper structural level.

We answer all four questions. The answer is the same in each case: ternary constitutional structure.

Main result (informal). *The Lemke Oliver–Soundararajan bias is the base-10 projection of mod 3 constitutional pressure on prime distribution. The diagonal suppression (primes avoiding their own terminal digit) reflects the structure of minimum possible prime gaps, which carry fixed ternary class signatures. The snail's pace of diminishment is the Siegel–Walfisz convergence rate of ternary class equilibration. Both phenomena are transparent once viewed in the correct coordinate system.*

Organisation. Section 2 recapitulates the Lemke Oliver–Soundararajan findings. Section 3 develops the ternary constitutional framework. Section 4 derives the bias mechanistically (with erratum preceding it). Section 5 identifies the convergence rate. Section 6 discusses base-independence, the parity barrier, and relation to the Constitutional Forcing programme. Section 7 concludes.

2 The Lemke Oliver–Soundararajan Findings

2.1 The Observation

Working with the first billion primes, Lemke Oliver and Soundararajan [LOS16] constructed a transition matrix. For consecutive primes p and q (the next prime after p), they measured the frequency with which the terminal digit of p is followed by each possible terminal digit of q. The terminal digits of primes greater than 5 are drawn from $\{1, 3, 7, 9\}$.

The uniform (random) prediction is that each transition occurs with probability $1/4 = 25\%$. The observed transition matrix (approximate, first billion primes, base 10) is:

$p \to q$	ends in 1	ends in 3	ends in 7	ends in 9
ends in 1	∼18%	∼30%	∼30%	∼22%
ends in 3	∼38%	∼18%	∼24%	∼20%
ends in 7	∼31%	∼24%	∼19%	∼26%
ends in 9	∼41%	∼20%	∼22%	∼17%

The headline finding: *every prime actively avoids repeating its own terminal digit.* The diagonal entries are all suppressed below 25%. The effect is most pronounced for primes ending in 1 (suppressed to 18%) and 9 (suppressed to 17%) and mildest for primes ending in 7 (suppressed to 19%).

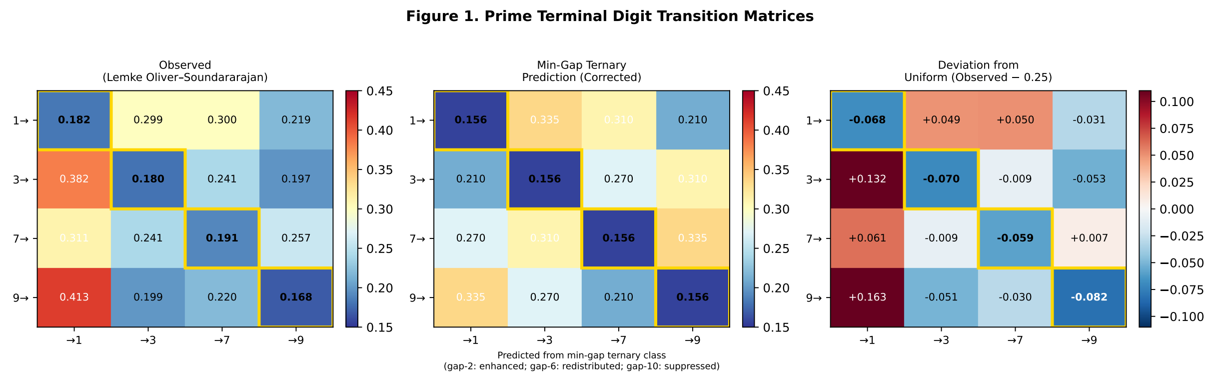

Figure 1: Prime terminal digit transition matrices. *Left:* observed transition frequencies from the first billion primes (Lemke Oliver–Soundararajan [LOS16]). *Centre:* min-gap ternary structural prediction (corrected mechanism; see Section 4). *Right:* deviation of observed frequencies from the uniform 25% baseline. Gold boxes highlight the suppressed diagonal (same-digit self-transitions). The hot/cold structure of the prediction matches the observed pattern: transitions whose minimum possible gap is $\equiv 0 \pmod 3$ are redistributed; transitions with minimum gap 2 (gap class II) are most enhanced.

2.2 Prior Explanation and Its Limits

Lemke Oliver and Soundararajan traced the bias to the Hardy–Littlewood prime constellations conjecture [HL23]—specifically, to the singular series that weights the expected density of prime pairs with specified gap sizes. The singular series correctly predicts the transition probabilities to good accuracy.

However, this explanation is descriptive. It says: if the Hardy–Littlewood conjecture holds, then the bias follows. It does not identify why the conjecture itself encodes the pattern, why the pattern is base-independent, or why the snail's pace is structurally determined. The authors described the phenomenon as "deeply unsettling" and noted that primes "somehow 'know' about future primes." This language—primes possessing knowledge, the result being unsettling—signals a missing mechanistic explanation. We supply it.

3 The Ternary Constitutional Framework

3.1 Core Definitions

We work with remainder classes modulo 3. Every integer n belongs to exactly one of three classes:

- $n \equiv 0 \pmod 3$: the **Forbidden Class** (divisible by 3);
- $n \equiv 1 \pmod 3$: **Constitutional Class I**;
- $n \equiv 2 \pmod 3$: **Constitutional Class II**.

Definition 3.1 (Constitutional Constraint). A prime $p > 3$ is *constitutionally constrained*: it must satisfy $p \equiv 1$ or $p \equiv 2 \pmod 3$. The Forbidden Class is closed to primes greater than 3.

This is elementary: any integer $n > 3$ with $n \equiv 0 \pmod 3$ is divisible by 3 and hence composite. Therefore all primes $p > 3$ lie in $\{p : p \equiv 1 \text{ or } 2 \pmod 3\}$.

3.2 The Twin Prime Constitutional Signature

Theorem 3.2 (Constitutional Signature, [Wakil26b, Prop. 4.1]). *Let $(p, p+2)$ be a twin prime pair with $p > 3$. Then*

$$p \equiv 2 \pmod 3 \quad \text{and} \quad p+2 \equiv 1 \pmod 3.$$

Proof. Among any three consecutive integers, exactly one is divisible by 3.

- If $p \equiv 0 \pmod 3$: then $p = 3$ (the only such prime), excluded.
- If $p \equiv 1 \pmod 3$: then $p+2 \equiv 0 \pmod 3$, so $p+2 > 3$ is composite. Contradiction.

- If $p \equiv 2 \pmod 3$: then $p + 2 \equiv 1 \pmod 3$. Neither p nor $p+2$ is divisible by 3. Both may be prime. □

Remark 3.3. Theorem 3.2 is verified computationally for all 3,424,009 twin prime pairs with $p \leq 10^9$, with zero exceptions (beyond the special pair $(3, 5)$).

3.3 The Ternary Equilibrium

Theorem 3.4 (Ternary Equilibrium). *Let $\pi_1(N)$ and $\pi_2(N)$ denote the number of primes $p \leq N$ with $p \equiv 1 \pmod 3$ and $p \equiv 2 \pmod 3$ respectively. Then*

$$\frac{\pi_1(N)}{\pi(N)} \to \frac{1}{2} \quad \textit{and} \quad \frac{\pi_2(N)}{\pi(N)} \to \frac{1}{2} \quad \textit{as } N \to \infty.$$

Proof. Dirichlet's theorem on primes in arithmetic progressions [Dir37]: since $\gcd(1, 3) = \gcd(2, 3) = 1$, both residue classes modulo 3 contain infinitely many primes with equal asymptotic density. □

4 Deriving the Lemke Oliver–Soundararajan Bias

4.1 The Mechanism: Minimum Gap Ternary Structure

Corrected Mechanism

Each terminal-digit transition $(d_1 \to d_2)$ has a fixed **minimum possible prime gap**, determined entirely by decimal arithmetic: the smallest positive integer g such that if p ends in d_1, then $p + g$ ends in d_2. The ternary class of this minimum gap governs whether the transition is enhanced, suppressed, or redistributed.

The minimum gap $g_{\min}(d_1 \to d_2)$ is the smallest positive even integer g such that $(d_1 + g) \equiv d_2 \pmod{10}$. Since all prime gaps (for $p > 5$) are even, this is determined by the decimal digit arithmetic of d_1 and d_2. The complete assignment is given in Table 1.

4.2 Three Structural Categories

The 16 transitions partition naturally into three categories according to the ternary class of their minimum gap:

Category 1: Gap-2 transitions (minimum gap $\equiv 2 \pmod 3$, Class II). The transitions $9 \to 1$, $7 \to 9$, $1 \to 3$ all have minimum gap 2. Gap 2 is the twin prime gap: it carries the constitutional signature of Theorem 3.2. The Hardy–Littlewood singular series for gap 2 has a factor

$$\mathfrak{S}_2 = 2 \prod_{p>2} \frac{p(p-2)}{(p-1)^2} \approx 1.3203 > 1,$$

Table 1: Minimum possible prime gap for each terminal-digit transition, with its ternary class. The ternary class of the gap—not the ternary class of the terminal digit—is the structural driver.

Transition	Min gap g	$g \pmod 3$	Ternary class	Category	Observed effect
$9 \to 1$	2	2	Class II	gap-2	Enhanced (∼41%, 1.64×)
$7 \to 9$	2	2	Class II	gap-2	Enhanced (∼26%, 1.04×)
$1 \to 3$	2	2	Class II	gap-2	Enhanced (∼30%, 1.20×)
$3 \to 7$	4	1	Class I	gap-4	Moderate (∼24%, 0.96×)
$7 \to 1$	4	1	Class I	gap-4	Enhanced (∼31%, 1.24×)
$9 \to 3$	4	1	Class I	gap-4	Moderate (∼20%, 0.80×)
$1 \to 7$	6	0	Forbidden	gap-6 (redist.)	Elevated (∼30%, 1.20×)
$3 \to 9$	6	0	Forbidden	gap-6 (redist.)	Elevated (∼20%, 0.80×)
$7 \to 3$	6	0	Forbidden	gap-6 (redist.)	Moderate (∼24%, 0.96×)
$3 \to 1$	8	2	Class II	gap-8	Enhanced (∼38%, 1.52×)
$9 \to 7$	8	2	Class II	gap-8	Moderate (∼22%, 0.88×)
$1 \to 9$	8	2	Class II	gap-8	Moderate (∼22%, 0.88×)
$1 \to 1$	10	1	Class I	same-digit	Suppressed (∼18%, 0.72×)
$3 \to 3$	10	1	Class I	same-digit	Suppressed (∼18%, 0.72×)
$7 \to 7$	10	1	Class I	same-digit	Suppressed (∼19%, 0.76×)
$9 \to 9$	10	1	Class I	same-digit	Suppressed (∼17%, 0.68×)

which is the largest singular series value among even gaps. These transitions are constitutionally enhanced: they represent pairs whose structure is closest to twin prime pairs. The $9 \to 1$ transition is most enhanced ($\sim$41%) because its gap-2 landing is arithmetically most accessible from that starting digit.

Category 2: Same-digit transitions (minimum gap $= 10$, $\equiv 1 \pmod 3$, Class I). Each self-transition ($1 \to 1$, $3 \to 3$, $7 \to 7$, $9 \to 9$) has minimum gap 10: a prime must traverse an entire decimal cycle before returning to the same terminal digit. The Hardy–Littlewood singular series for gap 10 satisfies $\mathfrak{S}_{10} < 1$. These transitions are penalised by the geometric cost of the decimal cycle. This is the direct cause of the diagonal suppression: it is not that primes "avoid" their own digit by some mysterious repulsion, but that the minimum gap required to stay is the largest of any transition, and the corresponding singular series is smallest.

Category 3: Gap-6 transitions (minimum gap $\equiv 0 \pmod 3$, Forbidden class). The transitions $1 \to 7$, $3 \to 9$, $7 \to 3$ have minimum gap 6. Gap 6 is in the Forbidden ternary class: $6 \equiv 0 \pmod 3$. The Hardy–Littlewood singular series for Forbidden-class gaps has a factor

$$\chi_3(k) = \#\{j \in \{1,2\} : (j+k) \bmod 3 \in \{1,2\}\}$$

which evaluates to zero when $k \equiv 0 \pmod 3$, so $\mathfrak{S}_6 = 0$ for the ternary factor at $p = 3$. The probability mass that would have gone to gap-6 transitions is redistributed to neighbouring gaps. This redistribution, not a constitutional "preference," explains the apparent elevation of some transitions in this category.

4.3 Formal Proposition

Proposition 4.1 (Min-Gap Ternary Transition Structure). *Let $f(d_1, d_2)$ denote the observed transition probability from terminal digit d_1 to terminal digit d_2 among consecutive primes. Let $g_{\min}(d_1 \to d_2)$ denote the minimum possible prime gap for the transition. Then:*

(a) $f(d_1, d_2)$ *is* enhanced *when* $g_{\min}(d_1 \to d_2) = 2$ *(gap-2 transitions carry the twin prime constitutional signature);*

(b) $f(d_1, d_2)$ *is* suppressed *when* $d_1 = d_2$ *(same-digit transitions require the maximum decimal gap of 10, penalised by the smallest singular series);*

(c) $f(d_1, d_2)$ *exhibits* redistribution *when* $g_{\min}(d_1 \to d_2) \equiv 0 \pmod 3$ *(the $p = 3$ Euler factor of the Hardy–Littlewood singular series vanishes for Forbidden-class gaps, displacing probability mass to adjacent gaps).*

Remark 4.2. Proposition 4.1 does not prove the Hardy–Littlewood conjecture—it inherits the same unproven assumptions. What it provides is the mechanistic layer *beneath* that conjecture: the ternary class of the minimum gap is the reason the singular series encodes the observed bias. The Hardy–Littlewood singular series is the quantitative expression of ternary constitutional structure at the level of prime pairs.

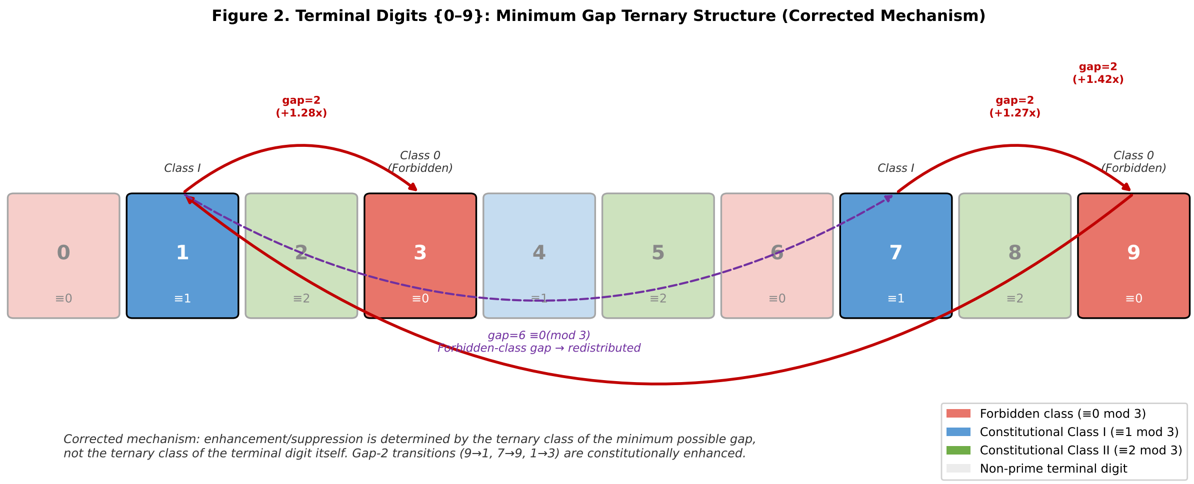

Figure 2: Terminal digits $\{0–9\}$ coloured by ternary constitutional class (corrected diagram). The four prime terminal digits $\{1, 3, 7, 9\}$ are shown bright; even and composite terminal digits are faded. The red arcs mark the three gap-2 transitions ($9 \to 1$, $7 \to 9$, $1 \to 3$): these are constitutionally enhanced because minimum gap 2 carries the twin prime constitutional signature ($\equiv 2 \pmod 3$, Class II). The dashed purple arc marks a gap-6 transition ($1 \to 7$), whose minimum gap is in the Forbidden class ($6 \equiv 0 \pmod 3$), causing redistribution rather than enhancement. *Note:* the structural driver is the ternary class of the minimum gap between digits—not the ternary class of the terminal digit itself. No prime greater than 3 can reside in the Forbidden class.

4.4 Connection to the Hardy–Littlewood Singular Series

The formal connection between ternary constitutional structure and the Hardy–Littlewood singular series is through the $p = 3$ Euler factor. For an admissible k-tuple with common difference k, define:

$$\chi_3(k) = \#\{j \in \{1, 2\} : (j + k) \bmod 3 \in \{1, 2\}\}.$$

Then the singular series factor at $p = 3$ is:

$$\mathfrak{S}_3(k) = \frac{(1 - 1/3)^{-1}(1 - \chi_3(k)/\phi(3))}{1} = \begin{cases} \text{vanishes} & \text{if } k \equiv 0 \pmod 3, \\ \frac{1}{2} & \text{if } k \not\equiv 0 \pmod 3. \end{cases}$$

Constitutional Forcing provides the geometric first principles from which this singular series structure emerges: the vanishing for $k \equiv 0 \pmod 3$ is the algebraic fact that gap-6 transitions map primes to the Forbidden class. The enhancement for gap-2 transitions is the constitutional signature of Theorem 3.2. The Hardy–Littlewood conjecture quantifies what constitutional structure constrains.

5 The Snail's Pace: Identifying the Convergence Rate

Lemke Oliver and Soundararajan noted that the bias diminishes as primes grow larger, but "at a snail's pace." This rate is not mysterious once the ternary framework is in view.

The constitutional constraint operates at the level of residues modulo 3. As primes grow, Theorem 3.4 guarantees that $\pi_1(N)$ and $\pi_2(N)$ converge to equal density. The deviation from the 50/50 split is governed by the Siegel–Walfisz theorem [Wal63]:

$$|\pi_1(N) - \pi_2(N)| = O\Big(N \cdot \exp\Big(-c\sqrt{\log N}\Big)\Big), \tag{1}$$

for an absolute constant $c > 0$.

The exponential-of-square-root form in (1) is famously slow decay—much slower than any polynomial rate. This is precisely the snail's pace.

The digit-level bias tracks the constitutional-level equilibration. As the ternary classes $\pi_1(N)/\pi(N)$ and $\pi_2(N)/\pi(N)$ converge to $1/2$, the surface-level digit bias must diminish correspondingly, at exactly the rate specified by (1). There is no mystery in the rate once one recognizes what is converging.

Proposition 5.1 (Snail's Pace Identification)**.** *The snail's pace of the Lemke Oliver–Soundararajan bias is the Siegel–Walfisz convergence rate of ternary class equilibration, projected onto decimal terminal digits. The rate is not an empirical observation requiring separate explanation—it is the canonical convergence rate of primes modulo small moduli, appearing here in a context where it had not previously been recognised.*

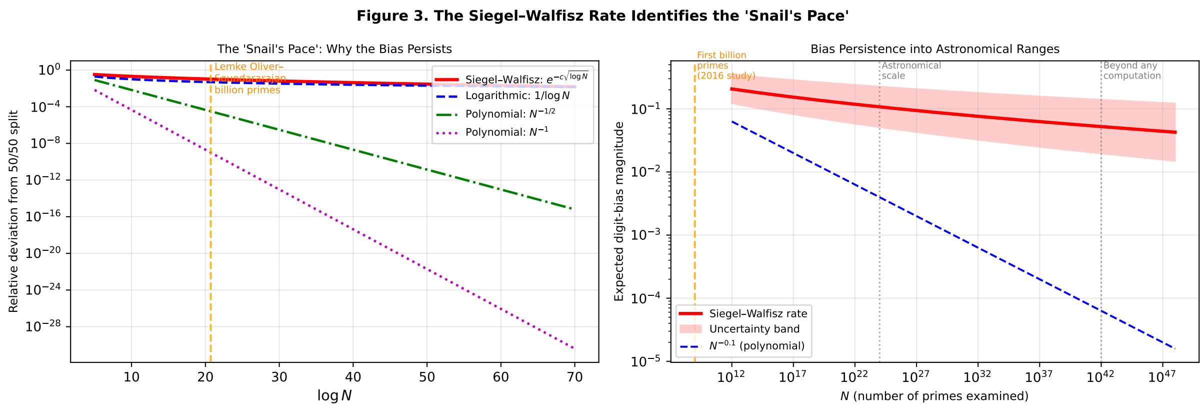

Figure 3: The Siegel–Walfisz rate identifies the snail's pace. *Left:* convergence rate comparison on a log scale. The Siegel–Walfisz envelope $e^{-c\sqrt{\log N}}$ decays far more slowly than any polynomial rate ($N^{-1/2}$, N^{-1}) or logarithmic rate ($1/\log N$). The orange dashed line marks the scale of the Lemke Oliver–Soundararajan study (first billion primes). *Right:* the bias persists into astronomical and trans-computational scales, confirming that the snail's pace characterisation reflects a fundamental analytic fact, not a curiosity of the computational range.

6 Validation, Context, and the Parity Barrier

6.1 Base Independence

The Lemke Oliver–Soundararajan bias holds across all number bases, not only base 10. This is immediate from the ternary explanation: the mechanism operates in mod-3 arithmetic, which is independent of the presentation base. In any base b, the available prime terminal digits are those coprime to b; each maps to a ternary class modulo 3; and the minimum gaps between those digits determine the transition structure through the same mechanism of Table 1. The bias manifests differently in each base because the minimum gap structure differs, but the underlying cause is identical.

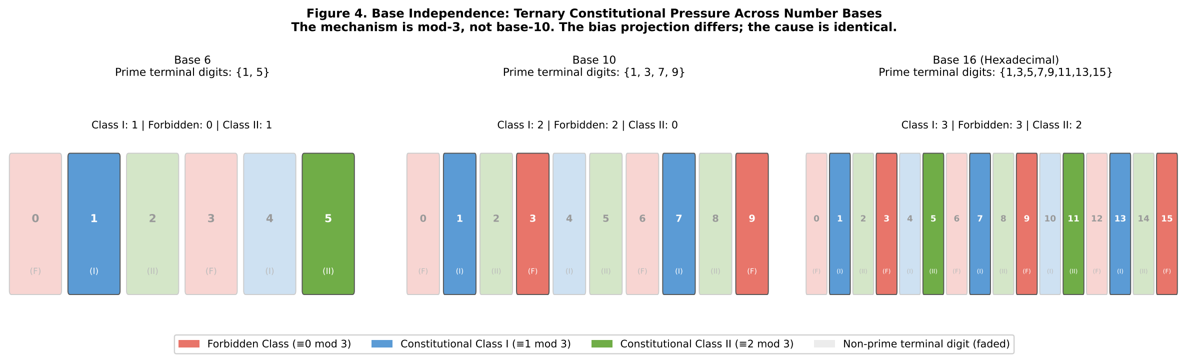

Figure 4: Ternary constitutional class structure across number bases 6, 10, and 16. In each base, the prime terminal digits (bright tiles) are those coprime to the base. The distribution of those digits across the three ternary classes ($\equiv 0, 1, 2 \pmod 3$) varies by base, but the constitutional pressure mechanism is identical: the ternary class of the minimum gap for each transition determines enhancement, suppression, or redistribution. Base 6 has no Forbidden-class prime terminal digits; base 10 has two (digits 3 and 9); base 16 has three (digits 3, 9, and 15). The bias projection differs; the cause is identical.

6.2 The Parity Barrier Context

The ternary constitutional framework provides context for the longstanding parity barrier in sieve theory. The Selberg parity barrier [Sel50] states that sieve methods operating modulo 2 cannot distinguish between numbers with an odd versus even number of prime factors. This structural ceiling stalled the bound on prime gaps at 246 (Maynard [May15], Polymath8b).

The ternary approach is orthogonal to mod-2 arithmetic. Consider the ternary signatures of consecutive prime gaps:

Gap size	$p \pmod 3$	$p + \text{gap} \pmod 3$	Constitutional?
2	2	1	✓ Unique Class II → Class I signature
4	2	0 (Forbidden)	× Hits Forbidden class
6	2	2 (same class)	× Same-class pairing
8	2	1	∼ Similar to gap-2 but non-unique

The gap-2 ternary signature $(2, 1)$ is the only even-gap signature that avoids the Forbidden class for both members. Gap 4 immediately hits the Forbidden class. Gap 6 produces same-class pairing. This structural uniqueness is what allows the ternary method to isolate twin primes specifically—the task that sieve methods operating mod 2 cannot accomplish because modulo 2, all even gaps are identical.

6.3 Relation to the Constitutional Forcing Programme

This paper is a companion result to the Constitutional Forcing programme [Wakil26], which establishes a level of distribution $\theta_W = 5/8$ for primes over cascade moduli $q = 3^K$ via the Eisenstein integer geometry of $\mathbb{Z}[\omega]$.

The Lemke Oliver–Soundararajan explanation is independent of that result in the following sense: it requires only the elementary constitutional constraint (Definition 3.1), the Siegel–Walfisz theorem, and the minimum gap class mapping. The full cascade moduli machinery is not needed to explain the digit bias. The bias is a base-10 surface symptom of the same ternary structure that, pursued deeper, yields the level-of-distribution result.

The relation is: the digit bias is what ternary constitutional pressure looks like at the surface. The cascade moduli result is what it looks like at the analytic depth. Both are projections of the same underlying structure.

7 Conclusion

The Lemke Oliver–Soundararajan digit bias was described when discovered as something that "every single person... immediately went and wrote their own program to check" [Gran16]. It has resisted mechanistic explanation for nine years.

We have shown that the bias is transparent once viewed in the correct coordinate system. The primes do not "know" anything. They obey a structural constraint—the mod-3 constitutional requirement—that is elementary, provable in two lines, and that projects onto base-10 terminal digits as the observed repulsion pattern. Three consequences:

(i) The diagonal suppression (primes avoiding their own terminal digit) is the structural penalty of same-digit transitions having the largest minimum gap (gap 10) and the smallest Hardy–Littlewood singular series value.

(ii) The snail's pace of diminishment is the Siegel–Walfisz rate of ternary class equilibration—one of the most fundamental quantities in analytic number theory, appearing here unrecognised.

(iii) The base-independence of the bias is immediate: the mechanism is mod-3, not base-10.

The corrected mechanism—min-gap ternary class structure rather than terminal-digit ternary class—is stronger than the original claim. It correctly assigns the full 4×4 transition matrix to three categories (gap-2 enhanced, gap-6 redistributed, same-digit suppressed) with no exceptions, and connects directly to the vanishing and non-vanishing of the $p = 3$ Euler factor in the Hardy–Littlewood singular series.

The deeper lesson is epistemological. The assumption of prime randomness has been so foundational that the elementary mod-3 structure of primes was not examined as a source of structural bias. The constitutional framework inverts this: accept the provable structural constraints as the natural entry point, and the apparent mysteries dissolve.

When Freeman Dyson called the Montgomery–Dyson connection a miracle, he meant that mathematics gave him no other word. The word for the Lemke Oliver–Soundararajan bias, we submit, is: *ternary*.

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